

18th CIRP Conference on Intelligent Computation in Manufacturing Engineering

Measuring Systems Engineering Transformation: A systematic literature review

Iris Graessler, Benedikt Grewe*

*Heinz Nixdorf Institut, Fuerstenallee 11, 33102 Paderborn, Germany** Corresponding author. Tel.: +49-5251-60-6229. E-mail address: benedikt.grewe@hni.upb.de

Abstract

Established companies are undertaking major transformation initiatives of their corporate structures and organisational forms to cope with the complexity during the engineering of cyber-physical production systems (CPPS). A frequently discussed issue is the measurability of this transformation progress. This paper conducts a systematic literature analysis of approaches regarding measurability of transformation and evaluates their application in the context of a systems engineering transformation. Measurement criteria are derived from the identified approaches, categorised, and finally evaluated by industry experts regarding their applicability. The categorised measurement criteria can be used to accurately measure the progress of a transformation process.

© 2025 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<https://creativecommons.org/licenses/by-nc-nd/4.0>)

Peer-review under responsibility of the scientific committee of the 18th CIRP Conference on Intelligent Computation in Manufacturing Engineering (CIRP ICME '24)

Keywords: Organizational Transformation, Systems Engineering, Measurement, Metrics, Organizational Change

1. Introduction

In a constantly changing business world, the continuous adaptation and further development of corporate structures and organizational forms is essential [1]. By proactively acting, companies can not only minimize potential risks, but also take advantage of opportunities for growth and innovation [1, 2]. Due to increasing technological challenges in interdisciplinary engineering and increased customer requirements, Systems Engineering is becoming an important engineering approach for developing complex systems including Cyber-Physical Production Systems (CPPS) [3, 4]. The introduction of Systems Engineering means a fundamental transformation for engineering units with an impact on established processes, methods, tools and structures.

Various challenges are observed when conducting organizational transformation, causing a large number of transformations to fail [5]: These include resistance from

employees, communication breakdowns, a lack of acceptance of new processes and uncertainty about the future direction [6]. A structured change management approach gives guidance to minimize those challenges [5]. Nevertheless common well known fundamental change management approaches ([6–10]) do not focus on Systems Engineering transformation in particular and measuring of transformation. Although, [11] describes a SE-specific change management approach, but no concrete program monitoring is provided. For that reason, this research aims at providing an overview of existing measurement criteria, including metrics and maturity models, for assessing organizational transformation and verifying their applicability in terms of Systems Engineering transformation.

2. Organizational development and transformation

Organizational development refers to the process of continuously adapting structures, processes and cultures of

organizations in response to constantly changing environmental conditions [9, 12]. In this paper, the terms change and transformation are used synonymously and refer to a fundamental change from an actual state to a target state through a planned and organized change process (in accordance with [13]).

An organizational transformation is a complex process in which fundamental changes are made to existing structures, processes, technologies or cultures of organizations in order to adapt to new market conditions, increase competitiveness or achieve long-term goals [13]. Change management refers to all measures for targeted management and control of transformation and focuses on reducing obstacles and risks [9]. For example, targeted strategies for communication, training and employee involvement can minimize resistance and create acceptance for change [9]. In addition, structured change management enables a smooth transition by focusing on clear goals, appropriate resources, and a coordinated process [1]. Overall change management helps to overcome obstacles that often accompany change and ensure successful implementation of new organizational structures and forms [1, 9]. Central models depict essential phases, challenges and focal points of a transformation process (cf. eight steps by KOTTER [6], five steps by KRÜGER [7], five action fields according to GRAESSLER [14], 7-S framework by MCKINSEY [8]).

Various approaches [1, 6, 9, 15, 16] identify key success factors for successful transformations. These include an assessment of current status of internal and external factors and stakeholders, a defined change vision and strategy, implementation of change measures and pursuit of a roadmap, as well as tracking and reporting of transformation progress.

In industrial practice, progress assessment and location determination in transformation processes are of great interest, as professional failures, escalating costs and expanding project durations need to be avoided. Also in literature various authors address the necessity of measuring transformation progress and performance [7, 9, 15–17]. By measuring the performance of transformation, it is possible to evaluate the achievement of the desired goals, adapt measures and actions as required and thus reduce the likelihood of transformation failing [5]. Transformation-specific maturity models also enable an assessment through objective and subjective evaluation levels in defined fields of action [18].

3. Scientific approach

This research is built on a seven-step scientific approach: The initial research problem is derived from industrial practice and supported by literature (1). A systematic literature review in accordance with PRISMA is conducted to identify current research results in field of measuring transformation (2). Measurement criteria are derived from the identified research reports (3) and clustered in 13 categories which are orientated in dimensions (4). Overlap and focus of existing research are analyzed, and due to the lack of research on measuring systems engineering transformation, the adaptability of identified research results to systems engineering transformation is reviewed and conducted (5). The adaptation is validated (6) and reflected through insights of industrial practice (7).

4. Transformation Measurement Criteria

First, a systematic literature review is conducted, to identify existing approaches on measuring organizational transformation. The identified reports are analyzed and measurement criteria are derived. In a next step the measurement criteria are categorized and evaluated in terms of their applicability on Systems Engineering Transformation. Finally, the applicability is validated in expert interviews.

4.1. Systematic literature review

The PRISMA guideline (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) is used as scientific approach for systematic literature search and meta-analysis of this research work (see PRISMA flow chart in figure 1). The use of PRISMA is intended to ensure that this literature search is systematic and thorough. [19]

Two different search strings were identified for this research (see table 1). The first search string is intended to identify systems engineering transformation measurement criteria. The second search string expands the scope to identify overall organizational transformation related measurement criteria.

Table 1. search strategy and strings

Id	search string
1	("metrics" OR "key performance indicators" OR "measurement" OR "measuring" OR "maturity" OR "indicator") AND ("transformation") AND ("systems engineering")
2	("metrics" OR "key performance indicators" OR "measurement" OR "measuring" OR "maturity" OR "indicator") AND ("organizational transformation" OR "organizational transformation" OR "enterprise transformation" OR "organisational change" OR "organizational change" OR "business transformation")

To ensure a broad range of results, four different scientific online libraries were analysed within this research: SCOPUS (search fields: title, abstract, keywords), Web of Science, IEEE Xplore and the University Paderborn library (search fields: all fields) Overall search sequences and databases 2.691 records are identified.

Table 2. hits by database

search string id	SCOPUS	Web of Science	Library University Paderborn	IEEE Xplore	Sum
1	n = 183	n = 76	n = 33	n = 397	n = 599
2	n = 1.809	n = 808	n = 181	n = 102	n = 2.092
	n = 1.992	n = 884	n = 214	n = 499	n = 2.691

According to PRISMA in a first step the identified records are filtered, and duplications are removed. Therefore, the titles of the identified 2.691 records have been screened and been assessed if the following selection criteria are matching:

It is clear from the title that ...

- ... a transformation process / organizational change is mainly discussed in the research.
- ... transformation performance is considered.
- ... measurement of a transformation is considered.

Based on this selection 64 records are identified for further abstract screening. These records were screened based on the abstracts and evaluated for their relevance in terms of content. The resulting 13 reports are briefly analysed on measurement criteria for transformation assessment.

4.2. Identification and analysis of measurement criteria

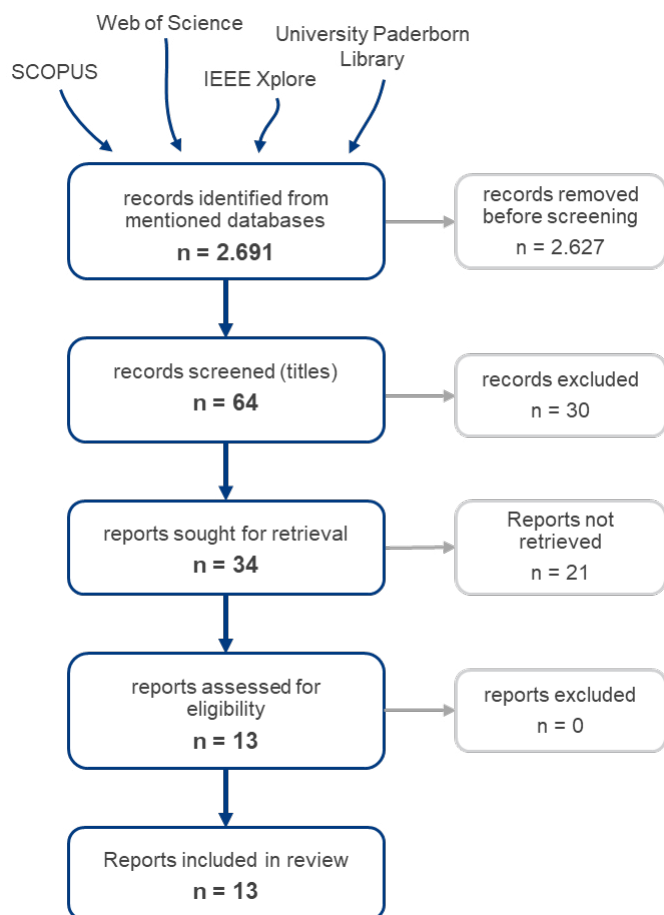


Fig. 1: PRISMA flow-chart of the executed literature review in this research

The literature review reveals 13 reports, which explicitly address transformation measurement, and define measurement criteria and metrics: BARTHEL as well as HARIYANTO ET AL. are conducting systematic literature reviews and reveal multiple studies on which success dimensions [20] and clusters with factors are derived [21]. CAMPANELLI ET AL. derive 23 success factors divided in six clusters for agile transformation [22]. ERRIDA ET AL. also proceed a literature review and identify literature based success metrics for change management [5]. DA ROSA and DA SILVA present a maturity model for business transformation and derive nine metrics based on the ISO 15504 [18]. HEIMICKE ET AL. built up a proceed for evaluation of implementation processes which is validated on three use cases [23]. LINES AND SMITHWICK as well as NASLUND AND NORRMAN conduct studies to analyse the implementation of change within companies [24] and the stakeholders view on organizational change [25]. NATHAN PERKINS ET AL. analyse 4 different assessment models on organizational transformation [26]. PETZOLT ET AL. analyse 148 items to measure digital transformations in SMEs based on early research results [27]. PIMENTEL AND MAJOR proceed a study for better understanding

of implementation of management processes and built up on the Common Assessment Framework 2020 for European quality management in public sector [28]. VICTORIA ET AL. examine the impact of organizational change on four defined key performance indicators [29]. Lastly, CAWSEY ET AL. present four different models for measuring organizational change [30].

The reports presented here offer different perspectives on different types of organizational transformation, but overall address different success factors and measurement criteria. In total, 216 measurement criteria are extracted from the 13 reports identified as relevant. The identified measurement criteria are consolidated on 98 transformation criteria, which are further analyzed. 13 categories of measurement criteria can be drawn into four broad dimensions (see figure 2).



Fig.2. Dimensions and categories of the identified measurement criteria

Starting from the internal organizational perspective, the dimensions of culture, processes, methods, tools and structures (PMTS), portfolio and governance are addressed, which are deeper introduced in the following. The analysis carried out shows that all reports have a common intersection of the measurement criteria presented in the central topics, in particular: organization, team, process, management and customer (see figure 3). In addition, the various approaches are characterized differently in the four derived dimensions. While the majority of the approaches set specific priorities in the dimensions, approaches by BARTHEL [20], ERIDA ET AL. [5], HARIYANTO ET AL. [21], NATHAN PERKINS ET AL. [26], PIMENTEL AND MAJOR [28] as well as CAWSEY ET AL. [30] offer



Fig. 3. Focal points of the identified references regarding considered measurement criteria

a broad overview of the dimensions without setting specific focal points. Therefore, Figure 4 shows the report-specific characteristics with respect to the previously introduced categories and dimensions.

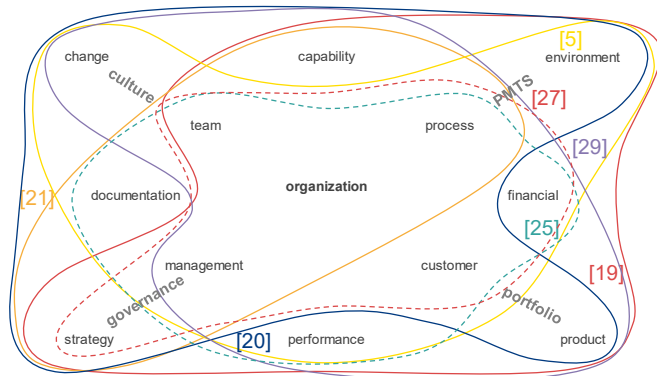


Fig. 4: Overview of the broad spread approaches

Further research shows specific focal point in different dimensions: PETZOLT ET AL. mainly focus measurement criteria regarding PMTS and product, while developing a digital transformation maturity model [27]. In this process they also identify 19 sub-dimensions for the evaluation of digital transformation, identified out of existing literature review [27]. CAMPANELLI ET AL. focus on criteria regarding governance, by especially clustering agile transformation success factors into six different groups [22]. The approaches by HEIMICKE ET AL. [23], LINES AND SMITHWICK [24] and NASLUND AND NORMAN [25] address mainly cultural aspects by analysing organizational change with qualitative measures and methods (see figure 5). These usually include retrospective surveys with various Likert-like scales as measures. Figure 5 shows the report-specific characteristics in the dimensions and categories mentioned above. Therefore, the highlighted arrows underline the individual focal dimensions of the identified reports.

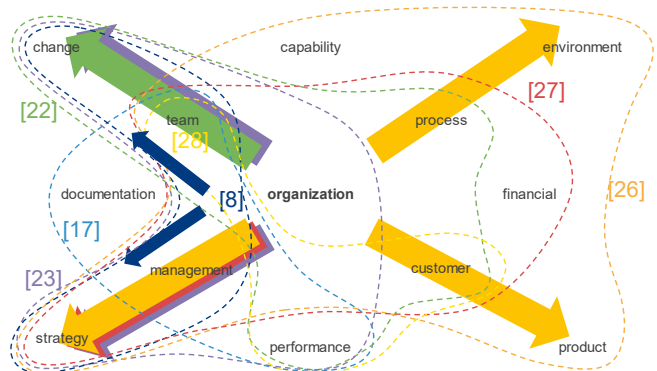


Fig. 5. Specialization of the approaches in direction of the prior identified dimensions

4.3. Categories of Measurement Criteria

Overall, most of the identified reports focus aspects of the internal organization perspective. Under this central pivot cluster in criteria like availability of resources, enabling infrastructure and criteria like structure or workplace quality are grouped, which form the basis for dimension specific deepening. Especially HARIYANTO ET AL. focus centralized aspects of the internal organization perspective: three of the four presented categories focus organizational, infrastructural

and workforce factors comprehensively [21], but also give slightly deepening in all four dimensions identified above.

The dimension PMTS contains the categories *process* and *environment*. Processes include quality or compatibility metrics but also metrics for toolsets which refer to PMTS due to enabling processes or the application of methods. External factors contain influences from outside of the organization, such as political regulations or cooperations with partners and suppliers. DA ROSA and DA SILVA derive therefore nine metrics of process assessment on basis of ISO 15504 [18]. In addition to the process perspective, ERIDA ET AL. explicitly focus on external measurability criteria regarding stakeholder perspective [18].

The dimension portfolio contains the categories *customer* and *product*. This dimension transitions from the involvement and satisfaction of customers to the availability and innovation of products and services. Especially PETZOLT ET AL. [27] and CAWSEY ET AL. focus in depth on customer and product perspective of transformation and offer therefore multiple measurement criteria and models, such as strategy maps, balanced score card and risk exposure calculator [30].

The dimension governance contains the categories *management* and *strategy*. Focus of this dimension are metrics concerning the management style and leadership engagement [25]. The outer area contains information which builds the basis for management decisions, such as strategic planning and business goals [27, 28].

The dimension culture contains the categories *team* and *change*. On the one hand employee roles, mindset and organization of teams is assessed in a variety of factors [24]. On the other hand change aspects like common sense of urgency and understanding of change, as well as organizational readiness for change and defined scope of action are considered [23, 24].

The cross-sectional cluster contain the categories *capability* [20–22, 27, 29, 30], *finance* [5, 20, 26, 27, 30], *performance* [5, 18, 20, 23, 24, 26, 29, 30] and *documentation* [5, 18, 21, 22, 26], which are sporadically addressed in particular research. Finance and performance metrics have similar output-oriented aspects. Criteria clustered around performance have value increasing factors, like company value and time saving [5, 29], while financial factors focus on monetary aspects, such as profitability and return on invest [5, 20]. Intellectual property clusters criteria surrounding knowledge and competences generated by team and processes. Lastly, the documentation cluster contains criteria that focus on monitoring, and change and process documentation aspects [22].

4.4. Transferability to systems engineering transformation

As literature research has shown, current approaches on different types of transformation exist, but focused approaches on measuring Systems Engineering transformation are missing. For that reason, the practical application and transferability of the identified criteria in context of Systems Engineering transformation is reviewed. In order to demonstrate the transferability of the identified approaches to Systems Engineering transformation, each criterion is evaluated individually with its intended guiding question. First, the identified criteria are evaluated in terms of their scope

regarding a focus on transformation program / controlling or content design of a company-specific Systems Engineering approach. Next, the type of assessment (quantitative vs. qualitative) is identified and the guiding questions are adapted, if possible, in the context of the Systems Engineering transformation. The assessment and adaption of the criteria is based on the authors' knowledge from several decades of consulting in the Systems Engineering transformation. In addition, experience from more than five different projects at automotive manufacturers and suppliers (level one to two) in the field of (model-based) Systems Engineering transformation and the applicability of A-SPICE served as input.

Out of the 87 consolidated criteria derived from the literature, 69 are identified as applicable to Systems Engineering transformation. 18 criteria are excluded, due to individual transformation-specific focal points that cannot be adapted or are not relevant to Systems Engineering transformation, such as political stability, telecommunications service quality or digital business models.

In the following table the applicability and focus of the literature-based criteria in field of systems engineering transformation is displayed.

4.5. Validation

In order to validate the derived measurement criteria of Systems Engineering transformation, five expert interviews were conducted with R&D consultants (consultant to manager level) as well as industry experts with more than a decade of experience in applying Systems Engineering and A-SPICE. In one-on-one interviews, the background of the research was presented and the transferred criteria, including related guiding questions, were reflected and discussed.

Although the experts initially agreed with large parts of the assessment, individual assessments were discussed in greater depth, placed in industrial context, and re-evaluated. Through discussion and classification of the criteria, an overall consensus was reached so that the list of applicable factors could be confirmed by all experts. Specific criteria of in-depth discussions are supplier performance, customer satisfaction, revenue per customer, agile champions, profitability.

With regard to supplier performance, the interfaces of engineering units were discussed, thus establishing the relevance of this factor in terms of systems engineering transformation. Customer satisfaction and revenue per customer represent both sides: the interest of the customer (customer satisfaction) and the interest of the company (revenue per customer). Therefore, both factors are identified as relevant. The aspect of agile champions was discussed in the context of "lighthouse projects" with high visibility for the organization, for which it is also considered relevant. Finally, the experts agree that profitability needs to be reviewed at the project level, but not at the enterprise level.

Experts pointed out that in addition to performance evaluation measuring of transformation (process perspective) the operational impact (transformation product perspective), regarding e.g. engineering performance increase, better quality, higher innovation level, is highly requested in industrial field.

In addition, experts highlighted the interfaces and overlapping of derived categories, for further analysis. The experts confirmed the scope and focus of the identified factors, but pointed out that each factor needs to be specified in the context of Systems Engineering transformation.

Table 3. Applicability and focus of identified criteria on Systems Engineering transformation

Dimension	Factor from literature	Applicability in SE Transformation
organization internal	collaboration	yes
	governance structure	yes
	IT security	no
	...	...
governance	leadership commitment	yes
	vision	yes
	agile champions	yes
	...	...
culture	mindset and roles	yes
	coaching and mentoring	yes
	self-organized teams	yes
	...	...
PMTS	external environment	yes
	process optimization	yes
	...	...
portfolio	company reputation	no
	...	...
	measurement / monitoring	yes
cross-sectional	Documentation	yes
	...	...

5. Discussion

Within this research, a large number of currently applied transformation measurement criteria is identified. For industrial application, the criteria are categorized, and relevant dimensions of transformation measurement are derived. The derived categories and dimensions provide an orientation and framework for industrial tailoring and application. For industrial application, companies need to identify criteria in all dimensions to get an overall orientation of their current progress and performance of transformation.

However, the mapping of criteria shows that the literature-based criteria provide a broad basis of relevant focal points but must be specified for use in the context of Systems Engineering transformation. Based on insights from industrial practice, available datapoints and artefacts need to be identified to achieve a specified and complete set of criteria. It is noticeable that surveys in terms of measuring transformation are predominantly carried out in guided interviews with questionnaires using Likert scales. Thereby a major part of criteria is described in a qualitative way, which should be quantified to achieve a unique assessment basis.

It should also be emphasized that the majority of identified approaches are written with a business background and thus focus strongly on a processual view. However, for a meaningful application in Systems Engineering transformation, the operationalization view, with a focus on actual change in CPPS engineering, is relevant in addition to the process view.

6. Conclusion and further research

Current approaches on Systems Engineering transformation do not focus on measurement aspects. Therefore, this research identifies common measurement criteria in organizational transformation and is reviewing their applicability in Systems Engineering transformation in terms of engineering CPPS.

Within this research, 69 criteria have been identified as applicable for measuring the progress of Systems Engineering transformation. Expert interviews and the authors' experience prove the applicability of the selected measurement criteria for Systems Engineering transformation but point out necessary specification, especially regarding the operational engineering view. Therefore, CPPS engineers must be involved in further specification of measurement criteria.

With the large number of criteria identified and an initial classification in context of Systems Engineering transformation, this research forms a basis for further research and application in industrial practice. Further research to derive criteria from industrial practice as well as quantification and relation to key performance indicators out of Systems Engineering transformation context is required. In addition, established Systems Engineering maturity models need to be identified and combined with the identified criteria into an overarching System Engineering transformation performance model.

References

- [1] Burnes B. Managing change: A strategic approach to organisational dynamics. 2. ed. London u.a.: Pitman 1996.
- [2] Macpherson C. The change catalyst: Secrets to successful and sustainable business change. Chichester, West Sussex, United Kingdom: Wiley 2017.
- [3] Kübler K, Scheifele S, Scheifele C, Riedel O. Model-Based Systems Engineering for Machine Tools and Production Systems (Model-Based Production Engineering). *Procedia Manufacturing* 2018; 24: 216–21
- [4] Feldmann S, Kernschmidt K, Wimmer M, Vogel-Heuser B. Managing inter-model inconsistencies in model-based systems engineering: Application in automated production systems engineering. *Journal of Systems and Software* 2019; 153: 105–34
- [5] Errida A, Lotfi B, Semma E. Measuring change management performance: A case study of a Moroccan construction company. In: *Measuring change management performance: A case study of a Moroccan construction company*; 2018; 206–13.
- [6] Kotter JP. Leading change. Brighton: Harvard Business Review Press 2012.
- [7] Krüger W, Bach N. Excellence in Change. Wiesbaden: Gabler Verlag 2014.
- [8] Joseph Galli B. Change Management Models: A Comparative Analysis and Concerns. *IEEE Eng. Manag. Rev.* 2018; 46(3): 124–32
- [9] Hartwich E. Grundlagen Change Management: Organisationen strategisch ausrichten und zur Exzellenz führen. 1. Auflage. Stuttgart/München: Richard Boorberg Verlag GmbH & Co KG 2014.
- [10] Burnes B. Kurt Lewin and the Planned Approach to Change: A Re-appraisal. *J Management Studies* 2004; 41(6): 977–1002
- [11] Gräßler I, Oleff C. Systems Engineering: verstehen und industriell umsetzen. Berlin, Heidelberg: Springer Berlin Heidelberg 2022.
- [12] Cummings TG, Worley CG. Organization development and change. 9th ed. Australia: South-Western Gengage Learning 2009.
- [13] Bittner-Fesseler A, Krutzke A, Hermann K. Change Kommunikation als Managementaufgabe. Wiesbaden: Springer Fachmedien Wiesbaden 2023.
- [14] Gräßler I. Kundenindividuelle Massenproduktion. Berlin, Heidelberg: Springer Berlin Heidelberg 2004.
- [15] Oakland JS, Tanner S. Successful Change Management. *Total Quality Management & Business Excellence* 2007; 18(1-2): 1–19 [https://doi.org/10.1080/14783360601042890]
- [16] the Association of Change Management Professionals. Change Management Standard© und ACMP Change Management Ethik-Kodex: Deutsche Übersetzung. Frankfurt am Main. Available from: URL: https://cdn.ymaws.com/www.acmpglobal.org/resource/resmgr/pdf/standards-german.pdf.
- [17] Horlick A. Metrics for Measuring Change Management: Prosci; 2024 [cited 2024 April 20] Available from: URL: https://www.prosci.com/blog/metrics-for-measuring-change-management#:~:text=We%20measure%20the%20outcomes%20by,applying%20the%20change%20(how%20well).
- [18] Da Rosa I, Da Silva MM. A Maturity Model for Business Transformation Management. In: *A Maturity Model for Business Transformation Management*; 2015; 60–7.
- [19] Page MJ, Moher D, Bossuyt PM, et al. PRISMA 2020 explanation and elaboration: updated guidance and exemplars for reporting systematic reviews. *BMJ* 2021; 372: n160 [https://doi.org/10.1136/bmj.n160][PMID: 33781993]
- [20] Barthel P. What is Meant by Digital Transformation Success? Investigating the Notion in IS Literature. In: Ahlemann F, Schütte R, Stieglitz S, editors. *What is Meant by Digital Transformation Success? Investigating the Notion in IS Literature*; 2021; 167–82.
- [21] Hariyanto A, Anggara SM, Kurniawan NB. Digital Business Transformation Framework Requirements: Study Literature. In: *Digital Business Transformation Framework Requirements: Study Literature*; 2022; 289–94.
- [22] Campanelli AS, Bassi D, Parreiras FS. Agile transformation success factors: A practitioner's survey. In: *Agile transformation success factors: A practitioner's survey*; 2017; 364–79.
- [23] Heimicke J, Czech C, Schoeck M, Mueller J, Rapp S, Albers A. Introducing Agility into the Processes of Manufacturing Companies: A Method for Evaluating Success, Support and Applicability. In: *Introducing Agility into the Processes of Manufacturing Companies: A Method for Evaluating Success, Support and Applicability*; 2022; 2463–72.
- [24] Lines BC, Smithwick JB. Best practices for organizational change management within electrical contractors. *International Journal of Construction Education and Research* 2019; 15(2): 136–59
- [25] Naslund D, Norrman A. A performance measurement system for change initiatives: An action research study from design to evaluation. *Business Process Management Journal* 2019; 25(7): 1647–72
- [26] Nathan Perkins L, Valerdi R, Nightingale D, Rifkin S. Organizational assessment models for enterprise transformation. In: *Organizational assessment models for enterprise transformation*; 2010; 809–23.
- [27] Petzolt S, Hölzle K, Kullik O, Gergeleit W, Radunski A. ORGANISATIONAL DIGITAL TRANSFORMATION OF SMEs—DEVELOPMENT AND APPLICATION OF A DIGITAL TRANSFORMATION MATURITY MODEL FOR BUSINESS MODEL TRANSFORMATION. *International Journal of Innovation Management* 2022; 26(3)
- [28] Pimentel L, Major MJ. Quality management and a balanced scorecard as supporting frameworks for a new management model and organisational change. *Total Quality Management and Business Excellence* 2014; 25(7-8): 763–75
- [29] Victoria B, Olena K, Olha M, Nataliia T, Denis S. Modeling sensitivity indices of industrial enterprise organizational change. *AMAZONIA INVESTIGA* 2021; 10(45): 230–41
- [30] Cawsey TF, Deszca G, Ingols C. Organizational change: An action-oriented toolkit. Third edition. Los Angeles, London, New Delhi, Singapore, Washington DC, Boston: Sage 2016.